

Data Visualization with Python

Objectives

- Create and modify plots using the matplotlib, seaborn, and pandas Python libraries
- Modify plot marker types and colors
- Identify the difference between MATLAB-style vs object oriented syntax for the Matplotlib library
- Add titles, labels, and limits to their plots
- Save plots to files
- Create subplots
- Identify the difference between plotting through Matplotlib, Seaborn, and Pandas libraries

Why use Python for Data Visualization

This lesson will give you the tools to be able to visualize your data. Data visualization is not just important to be able to explore and understand your data, but also to understand how your machine learning algorithm is performing. This lesson covers how to create plots using the matplotlib, seaborn, and pandas libraries. This includes confusion matrices, boxplots, and more. This is a topic that will be continuously reviewed throughout the course material.

Lesson Materials

Jupyter Notebooks and Datasets

Download and unzip this zip file: [Week1_DataVisualizationWithPython.zip](#)

<https://vetsintech.instructure.com/courses/241/files/45314?wrap=1> ↓

https://vetsintech.instructure.com/courses/241/files/45314/download?download_frd=1

I highly recommend that you have your own copy of each notebook running to follow along with the lesson. [There is also a YouTube video on these notebooks which you can reference later as well](#) [↪ \(https://youtu.be/OOLIVlleaN4\)](https://youtu.be/OOLIVlleaN4). In order, the notebooks used are:

1. BasicsMatplotlib.ipynb
2. MarkerTypeColor.ipynb
3. MATLABvsObjectSyntax.ipynb
4. TitlesLabelsLimits.ipynb
5. Grids.ipynb
6. Legends.ipynb
7. SavePlotsFile.ipynb
8. Subplots.ipynb
9. Heatmaps.ipynb

One question I often get is what is the difference between the plotting functionality of pandas, matplotlib, and seaborn. In result, I made a [video](https://youtu.be/BE8CVGJuftI?si=G-QkQDYW9tyBHKSt)  [\(https://youtu.be/BE8CVGJuftI?si=G-QkQDYW9tyBHKSt\)](https://youtu.be/BE8CVGJuftI?si=G-QkQDYW9tyBHKSt) and corresponding notebook which goes over this.

10. MatplotlibWrappersPandasSeaborn.ipynb

In the learn by doing portion of the lecture, **your task is to:**

1. Create a visualization comparing car interest payments on 3,5,7, and 10% fixed rate loans.
2. Create a visualization comparing home loan interest payments with 30 year term and 3,5,7, and 10% fixed rate loans.
3. (bonus hard question if you get bored) Assuming you got an 6% fixed 30 year home loan, and you refinance the loan 5 years into the loan at 5% fixed, what would be the difference in total interest payed between the loans.

Recommended Reading

[Visualization with Matplotlib \(an excerpt from the Python Data Science Handbook by Jake](#)

[VanderPlas\)](#)  [_ \(https://jakevdp.github.io/PythonDataScienceHandbook/04.00-introduction-to-](https://jakevdp.github.io/PythonDataScienceHandbook/04.00-introduction-to-matplotlib.html)

[matplotlib.html\)](https://jakevdp.github.io/PythonDataScienceHandbook/04.00-introduction-to-matplotlib.html)  [_ \(https://blog.dominodatalab.com/spark-dask-ray-choosing-the-right-framework\)](https://blog.dominodatalab.com/spark-dask-ray-choosing-the-right-framework)